

Dual Inline Memory Modules (DIMMs)

The DIMM is a newer memory module, intended for use in fifth- and sixth-generation computer systems. DIMMs are 168 pins in size, and provide memory 64 bits in width. They are a newer form factor and are becoming the de facto standard for new PCs. They are also not generally available in smaller sizes such as 1 MB or 4 MB.

Physically, DIMMs differ from SIMMs in an important way. SIMMs have contacts on either side of the circuit board but they are tied together. So a 30-pin SIMM has 30 contacts on each side of the circuit board, but each pair is connected. This gives some redundancy and allows for more forgiving connections since each pin has two pads.

This is also true of 72-pin SIMMs. DIMMs, however, have different connections on each side of the circuit board. So a 168-pin DIMM has 83 pads on each side and they are not redundant. This allows the packaging to be smaller, but makes DIMMs a bit more sensitive to correct insertion and good electrical contact.

DIMMs are inserted into special sockets on the motherboard, similar to those used for SIMMs. They are generally available in 8 MB, 16 MB, 32 MB and 64 MB sizes, with larger DIMMs also available. DIMMs are the memory format of choice for the newest memory technology, SDRAM.

DIMMs come in different flavours - 3.3 V and 5.0 V, and they come in either buffered or unbuffered versions. This yields a total of four different combinations. The standard today is the 3.3 volt unbuffered DIMM, and most machines use these.